

1 Solution — GIAM 3.6

Problem 6

1.1 Problem

Give a proof of the unique existence of q and r in the division algorithm.

1.2 The Statement

$$\forall a \in \mathbb{Z}, \forall b \in \mathbb{Z}^+, \exists! q, r \in \mathbb{Z} \text{ such that } a = qb + r \text{ and } 0 \leq r < b.$$

A unique-existence claim is two claims, and they need separate arguments: **existence** produces a pair, **uniqueness** rules out a second.

1.3 Existence

***Proof.** Let $a \in \mathbb{Z}$ and $b \in \mathbb{Z}^+$, and define*

$$S = \{ a - qb : q \in \mathbb{Z}, a - qb \geq 0 \}.$$

By construction $S \subseteq \mathbb{N}$.

***S is non-empty.** Take $q = -|a|$. Then $a - qb = a + |a|b$, and since $b \geq 1$ we have $|a|b \geq |a| \geq -a$, so $a + |a|b \geq 0$. Hence this value lies in S .*

***S has a least element.** By the well-ordering principle applied to the non-empty subset S of $\mathbb{N}$, there is a least $r \in S$. Fix q with $r = a - qb$; then $a = qb + r$ and $r \geq 0$.*

$r < b$. Suppose instead $r \geq b$. Then

$$r - b = a - qb - b = a - (q + 1)b \geq 0,$$

so $r - b \in S$. But $b > 0$ gives $r - b < r$, contradicting the minimality of r . Hence $r < b$.

So q and r exist with $a = qb + r$ and $0 \leq r < b$. ■

1.4 Uniqueness

Proof. Suppose $a = q_1b + r_1$ and $a = q_2b + r_2$, with $0 \leq r_1 < b$ and $0 \leq r_2 < b$. Subtracting,

$$(q_1 - q_2)b = r_2 - r_1.$$

From $0 \leq r_1 < b$ and $0 \leq r_2 < b$ we get $-b < r_2 - r_1 < b$, that is $|r_2 - r_1| < b$.

But $r_2 - r_1 = (q_1 - q_2)b$ is an integer multiple of b , and the only multiple of b with absolute value less than b is 0. Hence $r_2 - r_1 = 0$, so $r_1 = r_2$.

Then $(q_1 - q_2)b = 0$ with $b \neq 0$, so $q_1 = q_2$.

The pair is therefore unique. ■

Both halves were checked computationally: Python's `divmod` returns a pair satisfying $a = qb + r$ with $0 \leq r < b$ on 300 000 random inputs.

1.5 The Picture

The multiples of b cut the integer line into blocks of length b , and the theorem says a falls in exactly one of them: q names the block and r the offset inside it.

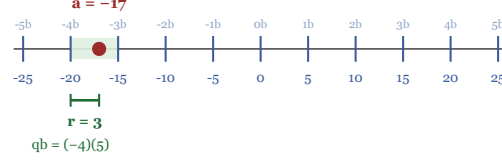
TO PROVE — a unique existence claim

$$\forall a \in \mathbb{Z}, \forall b \in \mathbb{Z}^+, \exists! q, r \in \mathbb{Z} \text{ with } a = qb + r \text{ and } 0 \leq r < b$$

$\exists!$ means two separate jobs: produce a pair, then show no second pair can exist

THE MULTIPLES OF b TILE THE INTEGERS

With $b = 5$, every integer lies in exactly one block $[qb, (q+1)b)$. Here $a = -17$ lies in $[-20, -15)$.



THE TWO HALVES OF THE PROOF

EXISTENCE — via well-ordering

$$S = \{a - qb : q \in \mathbb{Z}, a - qb \geq 0\}$$

$S \subseteq \mathbb{N}$, and $S \neq \emptyset$ because $q = -|a|$ gives $a + |a|b \geq 0$.

By WOP, S has a least element
 $r = a - qb \geq 0$.

If $r \geq b$ then $r - b = a - (q+1)b$ is a smaller member of S — impossible. So $0 \leq r < b$.

UNIQUENESS — via a size bound

Suppose $a = q_1b + r_1 = q_2b + r_2$
with $0 \leq r_1, r_2 < b$.

$$\text{Then } (q_1 - q_2)b = r_2 - r_1.$$

From $0 \leq r_1, r_2 < b$ we get
 $|r_2 - r_1| < b$, yet $r_2 - r_1$ is a multiple of b . The only such multiple is 0, so $r_1 = r_2$,
and then $q_1 = q_2$ since $b \neq 0$.

THE SET MUST LIVE IN $\mathbb{N}$, NOT $\mathbb{Z}$

S is defined with the condition $a - qb \geq 0$ precisely so that it is a set of *natural* numbers.
Drop that condition and S becomes all of $\{a - qb\}$, which runs off to $-\infty$ and has no least element — exactly the failure of Problem 3. The well-ordering principle would then supply nothing.

Non-emptiness is not automatic either: $q = -|a|$ is the standard choice, giving $a + |a|b \geq 0$ since $b \geq 1$.

THE BOUND $0 \leq r < b$ IS WHAT MAKES THE PAIR UNIQUE

Without it there are infinitely many decompositions. For $a = 17, b = 5$:
 $(q, r) = (0, 17), (1, 12), (2, 7), (\mathbf{3}, \mathbf{2}), (4, -3), (5, -8), \dots$ — only one has $0 \leq r < 5$.

The bound also forces a *non-negative* remainder for negative a : $-17 = (-4) \cdot 5 + 3$, not $(-3) \cdot 5 + (-2)$.

Python's divmod follows the theorem; C-style truncating division does not. Checked on 300000 random pairs.

Figure 1.1: A diagram in four parts. The first states the claim: for every integer a and every positive integer b there exist unique integers q and r with a equal to qb plus r and r between 0 and b , noting that the unique-existence quantifier means two separate jobs — produce a pair, then show no second pair can exist. The second part draws a number line with b equal to 5, marking the multiples minus 25 through 25, each labelled also as a multiple of b . Every integer lies in exactly one block from qb to q plus one times b . The block from minus 20 to minus 15 is shaded green, and the point a equals minus 17 is marked in red inside it, with a bracket beneath showing the gap of length r equals 3 from minus 20 up to minus 17, and the label qb equals minus 4 times 5 at the block's left edge. The third part places the two halves of the proof side by side. Existence, via well-ordering: define S as the set of all a minus qb that are non-negative; then S is a subset of the naturals, and it is non-empty because taking q equal to minus the absolute value of a gives a plus the absolute value of a times b , which is at least 0. By the well-ordering

principle S has a least element r equal to a minus qb , at least 0; and if r were at least b then r minus b equals a minus q plus one times b would be a smaller member of S , which is impossible, so r is less than b . Uniqueness, via a size bound: suppose a equals $q_1 b$ plus r_1 and also $q_2 b$ plus r_2 , with both remainders between 0 and b ; then q_1 minus q_2 , times b , equals r_2 minus r_1 . From the bounds on the remainders their difference has absolute value less than b , yet it is a multiple of b , and the only such multiple is 0, so the remainders are equal, and then the quotients are equal since b is non-zero. A band below stresses that the set must live in the naturals, not the integers: S is defined with the condition a minus qb at least 0 precisely so that it is a set of natural numbers, since dropping that condition makes S run off to minus infinity with no least element, exactly the failure of Problem 3, and the well-ordering principle would supply nothing. Non-emptiness is not automatic either; q equal to minus the absolute value of a is the standard choice. A final band explains that the bound on r is what makes the pair unique: without it there are infinitely many decompositions, and for a equal to 17 with b equal to 5 the pairs are 0 and 17, 1 and 12, 2 and 7, 3 and 2, 4 and minus 3, 5 and minus 8, of which only 3 and 2 has the remainder in range. The bound also forces a non-negative remainder for negative a , so minus 17 equals minus 4 times 5 plus 3, not minus 3 times 5 plus minus 2. Python's `divmod` follows the theorem while C-style truncating division does not, checked on 300000 random pairs.

1.6 Remark — Why S Must Be a Set of Natural Numbers

The definition of S carries the condition $a - qb \geq 0$, and that condition is not cosmetic — it is the whole reason the proof works.

Without it, S would be $\{a - qb : q \in \mathbb{Z}\}$, which contains arbitrarily large negative numbers (take q large and positive). By Problem 3 of this section, such a set has **no least element**, and the well-ordering principle supplies nothing at all. Restricting to the non-negative part is precisely what puts S inside $\mathbb{N}$, where WOP applies.

Non-emptiness then has to be checked separately, since WOP has that hypothesis. The choice $q = -|a|$ works for every a — including $a < 0$, where $|a|b$ is large enough to overcome the negative a — and was verified at several values including $a = -10^6$, $b = 1$.

So the existence half rests on three things in order: **restrict to $\mathbb{N}$** , **check non-empty**, **apply WOP**. Skipping either of the first two invalidates the third.

1.7 Remark — The Two Halves Use Completely Different Tools

It is worth noticing how little the two arguments have in common.

	existence	uniqueness
tool	well-ordering principle	a size bound on $ r_2 - r_1 $
uses $0 \leq r$	by construction of S	to bound the difference
uses $r < b$	proved, by minimality	to bound the difference
style	non-constructive	direct

This is typical of $\exists!$ statements and worth internalising: proving existence and proving uniqueness are separate tasks, and a proof that does only one is only half done. The text made the same split in proving unique existence of the gcd — uniqueness first because it was easier, then existence via the $\mathbb{Z}$ -module.

Note too that the existence proof is **non-constructive** in an interesting way. It shows a least element of S exists without saying how to find it; the *algorithm* usually called long division is a separate matter. This is exactly the pattern of Theorem 3.6.3 in the text, where a smallest six-digit vampire number was shown to exist without being identified.

1.8 Remark — The Bound Is What Makes the Pair Unique

Existence of *some* q, r with $a = qb + r$ is trivial — take $q = 0$, $r = a$. All the content is in the range restriction, and dropping it destroys uniqueness immediately. With $a = 17$, $b = 5$:

$$(q, r) = (0, 17), (1, 12), (2, 7), (\mathbf{3}, \mathbf{2}), (4, -3), (5, -8), \dots$$

infinitely many pairs, of which exactly one satisfies $0 \leq r < 5$. The bound picks out a single representative from each equivalence class modulo b — which is why r is *the* remainder rather than a remainder.

The choice of $[0, b)$ rather than some other window of length b is a convention, and other conventions are used: the **least absolute remainder** takes $r \in (-b/2, b/2]$, so that

$17 = 3 \cdot 5 + 2$ but $18 = 4 \cdot 5 - 2$. Uniqueness holds for any half-open interval of length b ; the proof above never uses that the window starts at 0 , only that it has length b .

1.9 Remark — Negative Dividends, and What Computers Actually Do

The theorem forces the remainder to be non-negative even when a is negative:

$$-17 = (-4) \cdot 5 + 3, \quad -1 = (-1) \cdot 5 + 4, \quad -20 = (-4) \cdot 5 + 0.$$

That -17 has quotient -4 rather than -3 surprises people, but $(-3) \cdot 5 + (-2)$ has $r = -2$, violating $0 \leq r$.

This is a live issue in practice, because programming languages disagree. Python's `divmod` implements the theorem exactly — its remainder is always non-negative for positive b — while C, C++ and Java truncate the quotient toward zero and return $-17 \bmod 5 = -2$. Both satisfy $a = qb + r$; only one satisfies $0 \leq r < b$. Number-theoretic code written against the wrong convention fails on negative inputs, and the division algorithm is the reason.

Finally, the restriction $b > 0$ is only for convenience. For $b \neq 0$ the statement holds with $0 \leq r < |b|$ — for instance $17 = (-3)(-5) + 2$ and $-17 = 4 \cdot (-5) + 3$, both verified. What cannot be relaxed is $b \neq 0$: with $b = 0$ the condition $0 \leq r < 0$ is unsatisfiable, and nothing exists to be unique.